

Mid-term examination on Logic Circuit Design (Group 1)

Christian Rinderknecht

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1 Balanced Ternary corrected

Question. Find a formula with a fixed number of operations for

$$S(q, n) = q^{n-1} + q^{n-2} + \cdots + q + 1, \quad \text{where } 1 < q.$$

What if $q = 2$ and $q = 3$?

Question. Prove that if the leftmost nonzero trit is 1, then the number is positive, otherwise it is negative.

2 Ternary

Question. Write an **efficient** C function which converts from ternary to decimal. The prototype is

```
int from3to10 (int* ternary, int t_size);
```

where `ternary` is an array of integers, of size `t_size`, representing a number in base 3. The most significant digit (also known as *trit*) is the first element of array `ternary`.

3 Conversions

Question. Decipher the secret message by converting **efficiently** from octal to hexadecimal the following numbers:

5413 3726355 12 157255 150320